BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 222 · Algorithms: synoptic exam challenge

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. True remaining cost is 4; is h(A)=7 admissible?

Q2. If g=3 and h=4, what is f?

Q3. What condition is needed for Dijkstra's standard shortest-path guarantee?

Q4. When partition sets the pivot at index 2, which recursive index ranges remain within indexes 0–4?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

partition — Rearrange around a pivot and place that pivot.

tentative distance — Current best-known cost from the start.

predecessor — Vertex immediately before another on a path.

heuristic — Estimate of remaining distance to the goal.

admissibility — A heuristic never exceeds the true cheapest remaining cost.

New learning

Match the algorithm to the task — Quicksort partitions an array. Dijkstra finds shortest routes on non-negative weighted graphs. A* adds a remaining-cost estimate.

Show all required intermediate states — For quicksort write arrays after partition. For shortest paths maintain distances, predecessors and a clear visited or frontier order.

Explain efficiency and preconditions — State when each algorithm is appropriate and how its method changes work.

Exam command words matter — Trace = show states. Explain = give a cause or reason. Compare = connect similarities and differences with evidence.

Worked example

Use last-element Lomuto partition on [5,2,4]. Give final pivot index.

1. Predict first — Start with pivot 4; 5 is too large to move to the smaller section.

2. Model response — Move 2 to the left: [2,5,4].

3. Model response — Swap pivot into index 1: [2,4,5]. Only one partition is being demonstrated.

We Do · Practise with guidance

Find P→T shortest route on P–Q=2, P–R=5, Q–R=1, Q–T=8, R–T=3.

```
Dijkstra: use the lowest total tentative distance.
```

Predict first

From P, Q gets 2 and R gets 5; choose Q.

Model response

From Q, R improves to 3 and T gets 10.

From R, T improves to 6. The shortest P→T path is P→Q→R→T, total cost 6.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Using last-element Lomuto quicksort on [7,2,6,1,5]: (a) state initial pivot [1]; (b) show the array after moving each qualifying value [3]; (c) give array and pivot index after partition [2]; (d) give fully sorted array [2]; (e) state typical average and worst-case time complexity [2].

Question 2

Find the shortest route A→E for non-negative edges: A–B=2, A–C=6, B–C=1, B–D=7, C–D=2, D–E=3, C–E=9. Give tentative-distance improvements [8] and final path [2].

Question 3

Graph edges S–A=2, S–B=5, A–B=1, B–G=3, A–G=7. Heuristics h(S)=5,h(A)=3,h(B)=2,h(G)=0. (a) Define g, h, f [3]; (b) show initial f for A and B [2]; (c) update B after expanding A [2]; (d) state shortest route and cost [2]; (e) evaluate h(A) admissibility [1].

Exit Ticket · Answer before checking

Quicksort, Dijkstra and A* are interchangeable because they all have a loop.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. No. 7 is an overestimate.

Q2. 7.

Q3. Non-negative edge weights.

Q4. 0–1 and 3–4.

You Do pages · Independent questions

Q1. MARK SCHEME /10: pivot 5 [1]; after moving 2 [2,7,6,1,5] [1], after moving 1 [2,1,6,7,5] [2] (no further value qualifies); pivot placed [2,1,5,7,6] [1], pivot index 2 [1]; fully sorted [1,2,5,6,7] [2]; average O(n log n) [1], worst $O(n^2)$ [1].

Q2. MARK SCHEME /10: after A, B2 and C6 [2]; settle B2, improve C3 and set D9 [2]; settle C3, improve D5 and set E12 [2]; settle D5, improve E8 [2]; shortest route A→B→C→D→E [1] costs 2+1+2+3=8 [1].

Q3. MARK SCHEME /10: g=cost travelled [1], h=estimated remaining cost [1], f=g+h [1]; after S, f(A)=2+3=5 [1], f(B)=5+2=7 [1]; after A, B improves to g3 [1], f(B)=3+2=5 [1]; S→A→B→G [1] cost 2+1+3=6 [1]; h(A)=3 is admissible because true remaining cost is 4 [1].

Exit Ticket · Model answer

False. They address different problems: ordering values, shortest paths and heuristic-guided search respectively.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.